



Jain College of Engineering **Dept. of Electronics and Communication Engineering**



Project Phase 1 Presentation
(Sub Code: BEC685) on

IoT and AI Powered Smart Agriculture Monitoring and Automation System

USN	NAME
2JI23EC157	Sushmita.Bambalwadi
2JI23EC158	Swapnil.Havale
2JI23EC160	Swati.Shivakale
2JI23EC182	Vishwas.Valmiki

Guided By,
Prof: Deepali.Patil
Assistant Professor



INDEX

O1 Introduction

O2 Literature
Survey

O3 Problem
Statement

O4 Objectives

O5 Applications

O6 Advantages



INTRODUCTION

- Agriculture is a vital sector that requires continuous monitoring to ensure healthy crop growth.
- Traditional farming methods depend heavily on manual inspection, which is time-consuming and less accurate.
- The integration of AI (Artificial Intelligence) and IoT (Internet of Things) can improve farming efficiency.
- This project focuses on smart agriculture monitoring using image-based analysis instead of physical sensors.
- A camera captures real-time images of crops.



- The captured images are processed using AI models for disease detection and health analysis.
- The system can identify early symptoms of crop diseases before they spread.
- It provides percentage-based health reports of crops (healthy vs affected area).
- The system can automatically trigger actions like alerts or irrigation control.
- This approach reduces dependency on expensive sensors and increases scalability.



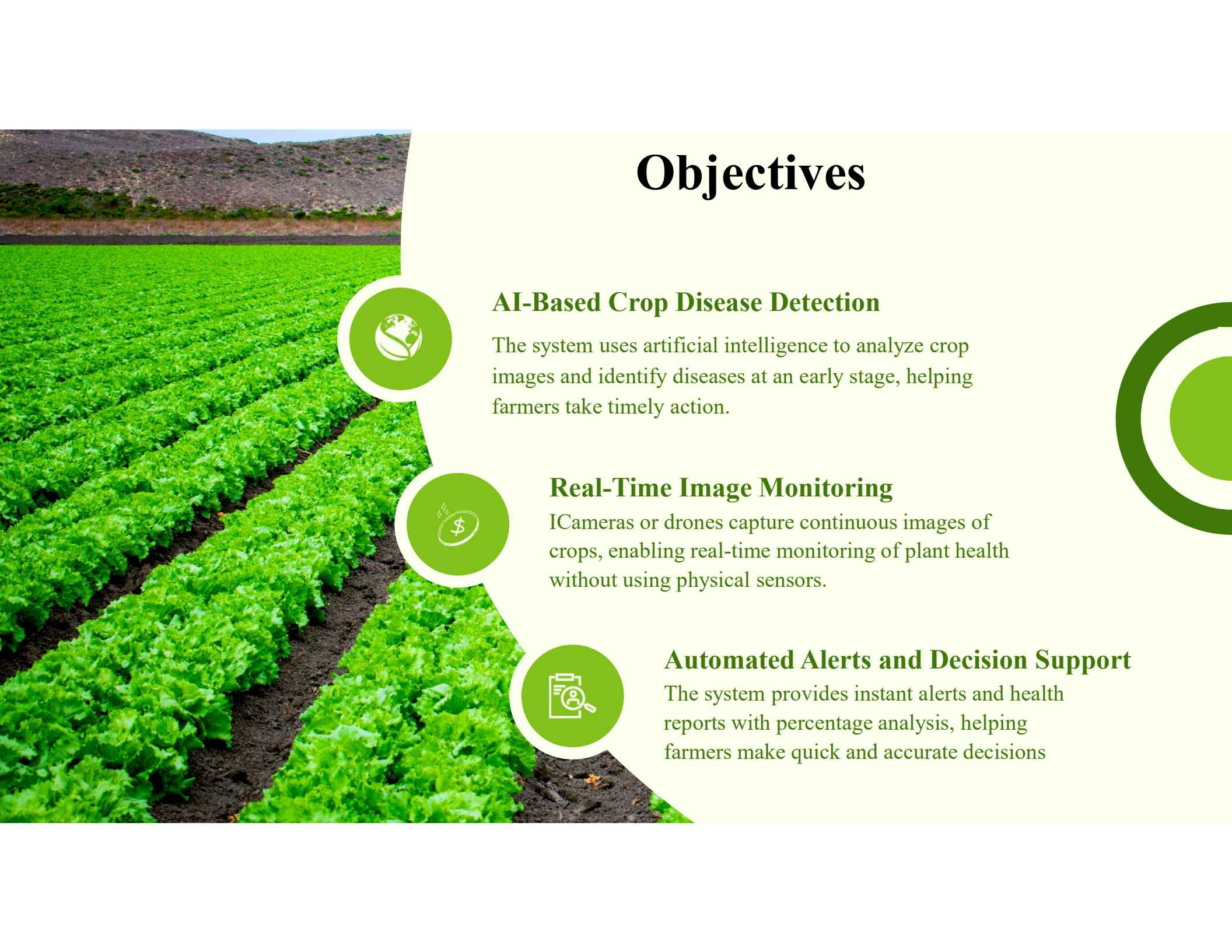
LITERATURE SURVEY

Author's Name and Year of Publication	Author's Contribution	Limitation	Proposed Solution
Krishna.G, Rajesh.B, Dinesh.B, Sravani.B, Rajesh.G, Dr. Prasad Dharnasi, Dr. M. Sarvanan(2026)	Developed an IoT and AI-based system to monitor soil and environmental conditions, enabling automatic irrigation and remote farm monitoring.	Limited AI accuracy, depends on internet, possible sensor errors, and lacks proper disease detection	Add advanced AI for prediction and disease detection, include SMS alerts, improve irrigation, and use solar power with a mobile app
Bilal Ahmad ¹ , Aleena Alam ¹ , Aiman Hamid ² , Muhammad Ameer Hamza ³ , Kashif Abbas ⁴ , Yunliang Ji ¹ , Naveed Abbas ⁵ , Sohail Abbas ¹ , Jamin Ali ¹ , Donato Romano ⁶ , Li Qiyun ^{1*} and Rizhao Chen(2026)	Developed an IoT-based smart farming system for soybean cultivation that monitors soil and environmental conditions and improves water management and crop productivity.	High cost, depends on power and internet, and limited to specific field conditions.	Use low-cost sensors, add AI for better automation, use solar power, and make it adaptable for different crops and fields.
Chi-Yong An ^{1,2} , Chulmin Song ¹ , Seokho Song ¹ , Seolmin Kim ¹ , Hoejeong Jeong ² , Jongsoo Ryu ² and Jinki Park ² (2026)	The authors developed a smart farming system using IoT sensors for soybean cultivation in former paddy fields. It monitors soil and environmental conditions and helps in efficient water management and improved crop growth.	The system is costly, depends on internet and power supply, and is mainly suitable only for specific field conditions.	Use low-cost sensors, add AI for better automation, integrate solar power, and make the system adaptable for different crops and environments.
J.Serin ¹ , G.Gifta Jerith ² , V.Ebenezer ^{3*} , K.Arul Jeyaraj ⁴ , A.Jenefa ³ , and M.Vargheese(2025)	The authors developed an AI-based smart farming system that uses image processing to detect plant health and nutrient deficiencies automatically and provide real-time monitoring.	Accuracy depends on image quality and environment, works on limited crops/datasets, and may be costly or complex for small farmers.	Use better datasets, advanced AI models, and integrate IoT with alert systems (like SMS/mobile) for accurate, real-time crop monitoring and quick farmer response.
S. P. Sudha ¹ · J. B. Shajilin Loret ^{1,2} (2025)	Reviewed ML and IoT-based techniques for crop monitoring, irrigation, and yield prediction to improve farming.	No practical implementation, high cost, needs technical knowledge, and depends on internet and data quality.	Develop low-cost systems, improve data accuracy, use advanced ML models, and provide farmer training.

PROBLEM STATEMENT

Farmers face difficulty in detecting crop diseases at an early stage using manual observation. Existing systems rely on sensors, which increase cost and complexity. There is a need for a cost-effective, image-based smart system that can monitor crop health and automate responses efficiently.





Objectives



AI-Based Crop Disease Detection

The system uses artificial intelligence to analyze crop images and identify diseases at an early stage, helping farmers take timely action.



Real-Time Image Monitoring

ICameras or drones capture continuous images of crops, enabling real-time monitoring of plant health without using physical sensors.



Automated Alerts and Decision Support

The system provides instant alerts and health reports with percentage analysis, helping farmers make quick and accurate decisions

Applications

Smart Farming

The system helps farmers monitor crop health efficiently using AI and improves overall farming productivity.

Early Disease Detection

It identifies crop diseases at an early stage, allowing farmers to prevent damage and reduce losses.



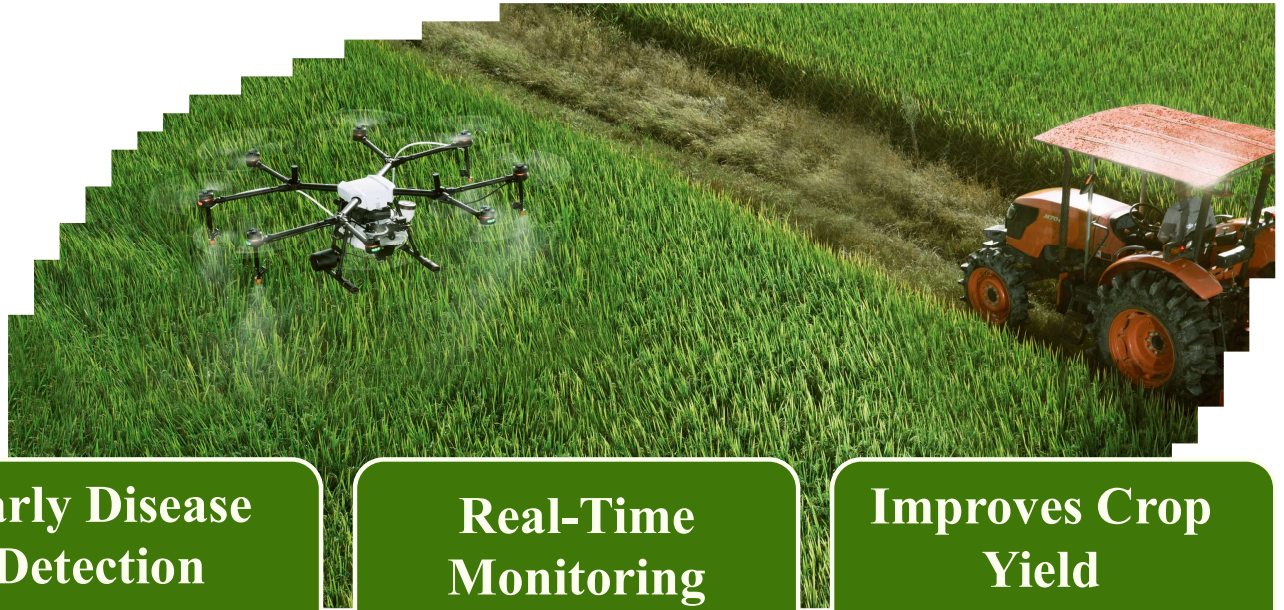
Yield Improvement

By detecting problems early and suggesting proper treatments, the system helps increase overall crop productivity.

Automated Farm Management

The system supports automation by sending alerts and helping in decision-making for irrigation and treatment.

Advantages



Low Cost

The system uses cameras and AI instead of expensive sensors, making it affordable and easy to implement for farmers.

Early Disease Detection

It detects diseases at an early stage before they spread, helping farmers take preventive measures and reduce crop loss.

Real-Time Monitoring

Image capture allows instant analysis of crop health, providing up-to-date information at any time.

Improves Crop Yield

By giving timely alerts and accurate insights, it helps farmers take proper actions, resulting in better crop growth and higher yield.

How to overcome the disadvantages



Improve Image Quality Issues

Use high-resolution cameras and proper lighting conditions to capture clear images, ensuring accurate AI analysis.



Increase Model Accuracy

Train the AI model with a large and diverse dataset of crop images to improve detection accuracy and reduce errors.



Power Backup Solution

Use batteries or solar panels to provide continuous power supply for cameras and devices in farms.

References

1. Krishna G., Rajesh B., Dinesh B., Sravani B., Rajesh G., P. Dharnasi, and M. Sarvanan, "Smart Agriculture System Using IoT with Help of AI-Techniques," International Journal of Computer Technology and Electronics Communication (IJCTEC), vol. 9, no. 2, pp. 479-487, Mar.-Apr. 2026.
<http://www.ijctece.com>
2. B. Ahmad et al., "Modern tools for sustainable agriculture: a review of intelligent crop protection technologies," Discover Agriculture, vol. 4, no. 19, 2026.
<https://doi.org/10.1007/s44279-025-00467-2>
3. C.-Y. An, C. Song, S. Song, S. Kim, H. Jeong, J. Ryu, and J. Park, "Development of an integrated smart farm system for open-field soybean cultivation in a former paddy fields," Front. Sustain. Food Syst., vol. 10, Art. no. 1702551, Jan. 2026, doi: 10.3389/fsufs.2026.
<https://doi.org/10.3389/fsufs.2026.1702551>
4. J. Serin et al., "AI-Driven Smart Farming for Automated Plant Health Monitoring and Nutrient Deficiency Detection," ASEAN Journal of Scientific and Technological Reports, vol. 29, no. 1, Dec. 2025, doi: 10.55164/ajstr.v29i1.258175.
<https://doi.org/10.55164/ajstr.v29i1.258175>
5. S. P. Sudha and J. B. Shajilin Loret, "A review on machine learning-based precision agriculture techniques for crop farming monitoring with IOT," Discover Environment, vol. 4, no. 10, Jan. 2026, doi: 10.1007/s44274-025-00305-8
<https://doi.org/10.1007/s44274-025-00305-8>

Thank You

